

# Buffer Stock Saving with Heterogeneous Agents

An example of the Econ-ARK MyST template

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*For everyone who waited for the solver to converge.*

*All models are wrong, but some are useful. George Box*

**Abstract** We solve a buffer stock saving model with the endogenous grid method and show how the Econ-ARK template typesets a working paper: title block, margin notes, equations, tables, code and author-year citations.

**Summary** Households save more when their income is uncertain. This paper shows how much more, and how quickly the extra saving can be computed.

**Keywords** consumption, precautionary saving, heterogeneous agents

**JEL codes** C63, D14, E21

## Materials

Run online  
Launch  
Starts in a few minutes

Code  
econ-ark/econ-ark-myst  
MIT license

REMARK  
BufferStockTheory

Source  
github.com/econ-ark/econ-ark-myst

Also as  
Latest version

## Key points

- A buffer of wealth protects households against bad income draws.
- The endogenous grid method solves the model without root finding.
- Patience, returns and risk set the size of the buffer.

## Working paper

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## 1 Introduction

Precautionary saving arises when income risk interacts with a convex marginal utility (Carroll 1997). The endogenous grid method (Carroll 2006) makes such models cheap to solve, because it replaces root finding with an inversion of the Euler equation.<sup>1</sup> Households hold a buffer of wealth against bad income draws. The size of that buffer depends on patience, the return on saving and the degree of risk.

## 2 Model

The consumer maximizes expected discounted utility subject to the budget constraint

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<sup>1</sup>The method inverts the first-order condition on a grid of end-of-period assets rather than solving it on a grid of market resources. `IndShockConsumerType` applies it to Section 2's model.

$$v(m_t) = \max_{c_t} u(c_t) + \beta \mathbb{E}_t[v(m_{t+1})], \quad m_{t+1} = R(m_t - c_t) + y_{t+1}. \quad (1)$$

The Euler equation implied by (1) is  $u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]$ , and with constant relative risk aversion  $u(c) = c^{1-\rho}/(1-\rho)$ .

## 2.1 Calibration

**Table 1** Baseline calibration.

| Parameter | Description            | Value |
|-----------|------------------------|-------|
| $\beta$   | Discount factor        | 0.96  |
| $R$       | Gross interest factor  | 1.03  |
| $\rho$    | Relative risk aversion | 2.00  |

### 2.1.1 Sources of the parameters

The discount factor and the interest factor follow [Carroll \(1997\)](#), and risk aversion is set to the value the literature uses most often.

## 3 Results

Table 1 lists the parameters used in the solution, which come from the 1990's literature. Proposition 1 describes the shape of the solution.

**Proposition 1** (Concavity). *If income risk is present, the consumption function is strictly concave. Its slope falls toward the perfect foresight marginal propensity to consume as wealth grows.*

*Proof.* See [Carroll \(1997\)](#) for the argument under constant relative risk aversion. □

#### Note

Admonitions take a rule in the palette rather than the filled box MyST draws by default. Econ-ARK blue carries this kind and important.

#### Tip

Green carries tip, hint and see also.

#### Warning

Orange carries attention, caution and this kind.

#### Danger

Pink carries this kind and error.

```

from HARK.ConsumptionSaving.ConsIndShockModel import
IndShockConsumerType

agent = IndShockConsumerType()
agent.solve()

```

Appendix A derives the Euler equation.

## Declarations

The authors report no competing interests. The authors used a large language model to draft the example text of this template and reviewed every sentence before release.

## Acknowledgments

We thank the Econ-ARK community for comments.

## Data availability

All code needed to reproduce the results is in the repository linked in the margin.

## References

- Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1): 1–56. <https://doi.org/10.1162/003355397555109>.
- Carroll, Christopher D. 2006. "The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems." *Economics Letters* 91 (3): 312–20. <https://doi.org/10.1016/j.econlet.2005.09.013>.

## Appendix A Derivation of the Euler equation

The first-order condition for  $c_t$  in (1) sets  $u'(c_t)$  equal to  $\beta R \mathbb{E}_t[v'(m_{t+1})]$ . The envelope condition gives  $v'(m_t) = u'(c_t)$ , and substituting it yields the Euler equation.